Дата-центр, бассейн, теплосеть: графики, показатели, стандартизация и юзкейсы

Инженерный аудиторский отчёт. 2025 год симуляции; проверка нормативных источников по состоянию на 23 октября 2026.

НЕ СЕРТИФИЦИРОВАНО. PUE=1,36 и ERF_proxy вычислены из синтетической нагрузки IT; оборудование, приборы, заводской COP и безопасная автоматика реального объекта не подтверждены.

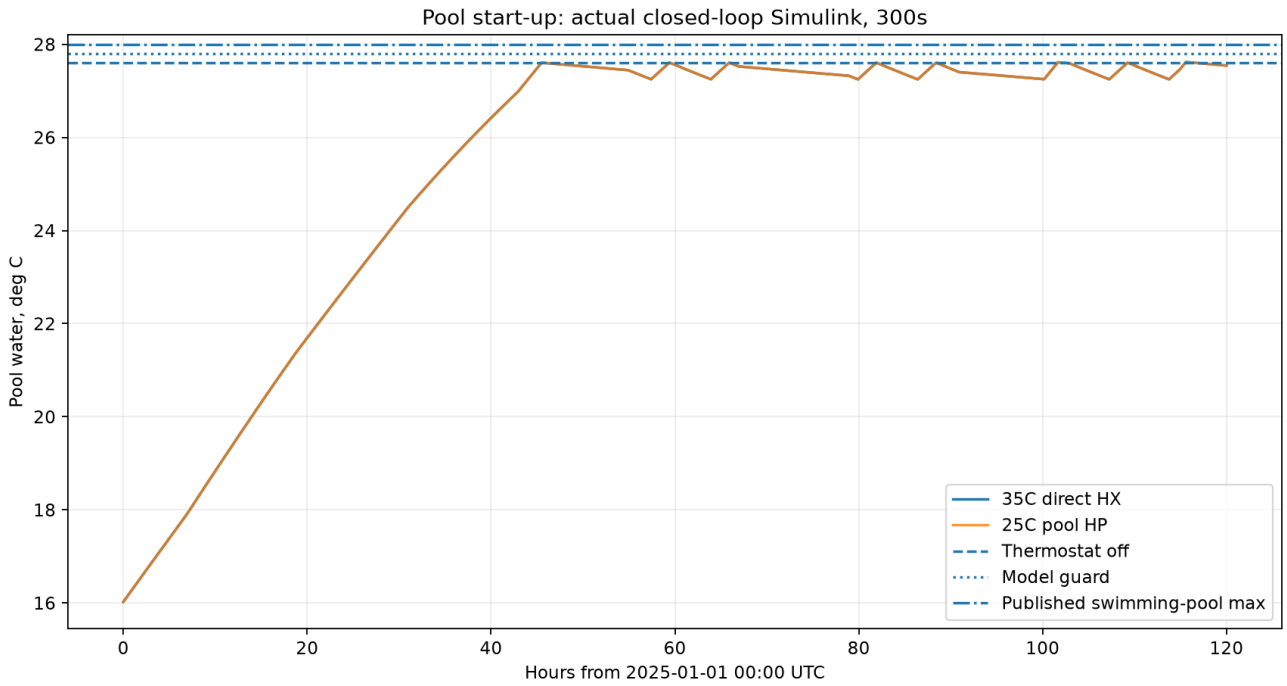
1. Проверенные ключевые показатели

Показатель	Значение	Единица	Статус / формула
E_IT	71,832.833	MWh/year	SIMULATED; IT thermal energy from assumed model power
E_recovered_source	38,274.674	MWh/year	SIMULATED; integral DCtoPool+DCtoDistrict
E_rejected_IT	33,558.159	MWh/year	SIMULATED; IT thermal energy minus source reused
Heat_pool_direct	250.125	MWh/year	SIMULATED; integral useful pool heat
Heat_district_HP35	47,769.079	MWh/year	SIMULATED; integral useful district HP output
Electric_district_HP35	9,744.530	MWh/year	SIMULATED; integral district HP compressor
Backup_gas_thermal	9,189.267	MWh/year	SIMULATED; integral useful backup gas heat
Pool_peak_temperature	27.620	degC	SIMULATED; max modeled pool water temp
Pool_hours_above_28	0.000	h	SIMULATED; temperature intervals exceeding 28C
Source_overallocation_peak	0.000	MW	QA; must be zero; total used source minus available
ERF_source_proxy	0.392	1	NONREPORTABLE; source heat extracted / assumed facility energy; NOT ERF
ERF_reportable	MISSING	1	MISSING_METER; boundary metered E_REUSE/E_DC
WUE_reportable	MISSING	m3/MWh	MISSING_METER; metered water consumption / IT energy

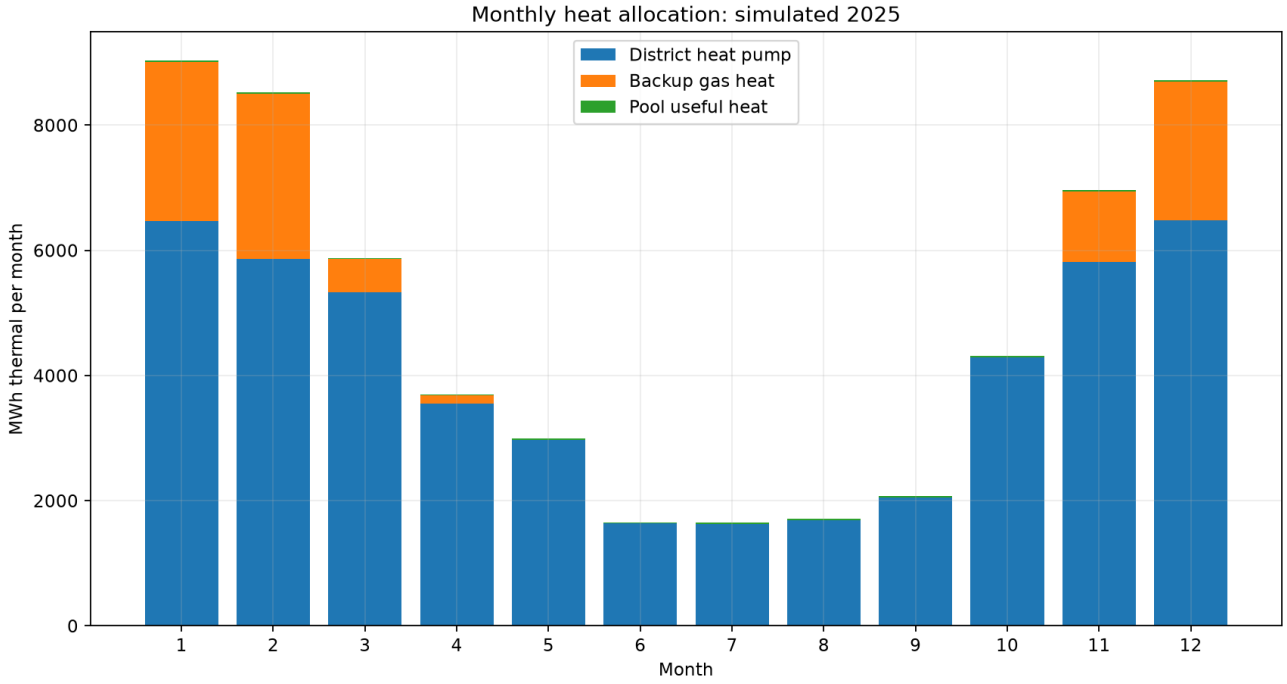
2. Графики реальных результатов симуляций

Полная галерея: Engineering_Dashboard_2026.html; каждый SVG приложен отдельно.

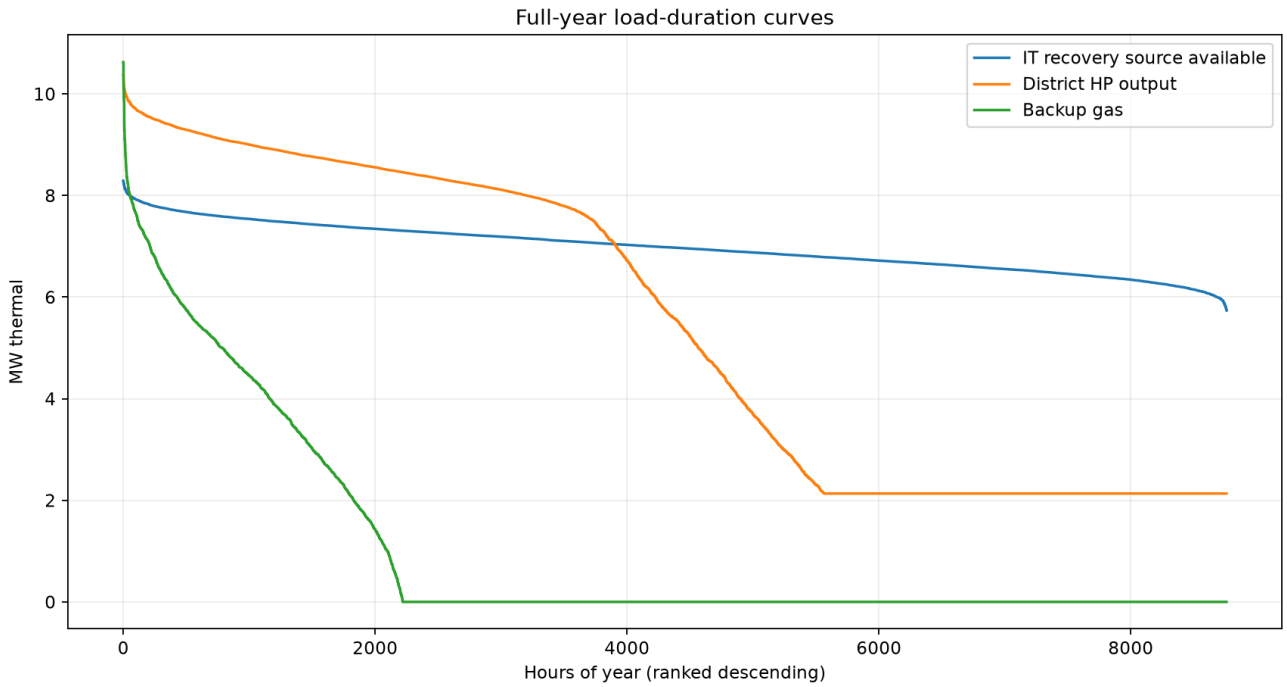
01 pool warmup



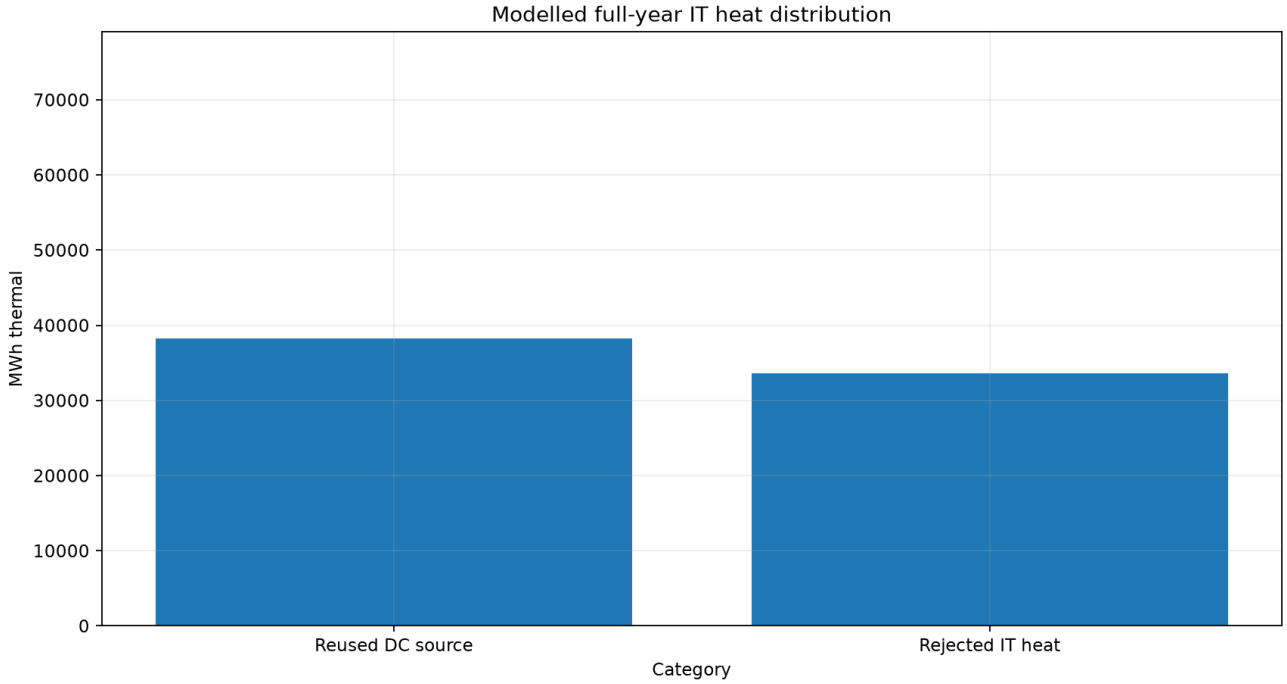
02 monthly energy



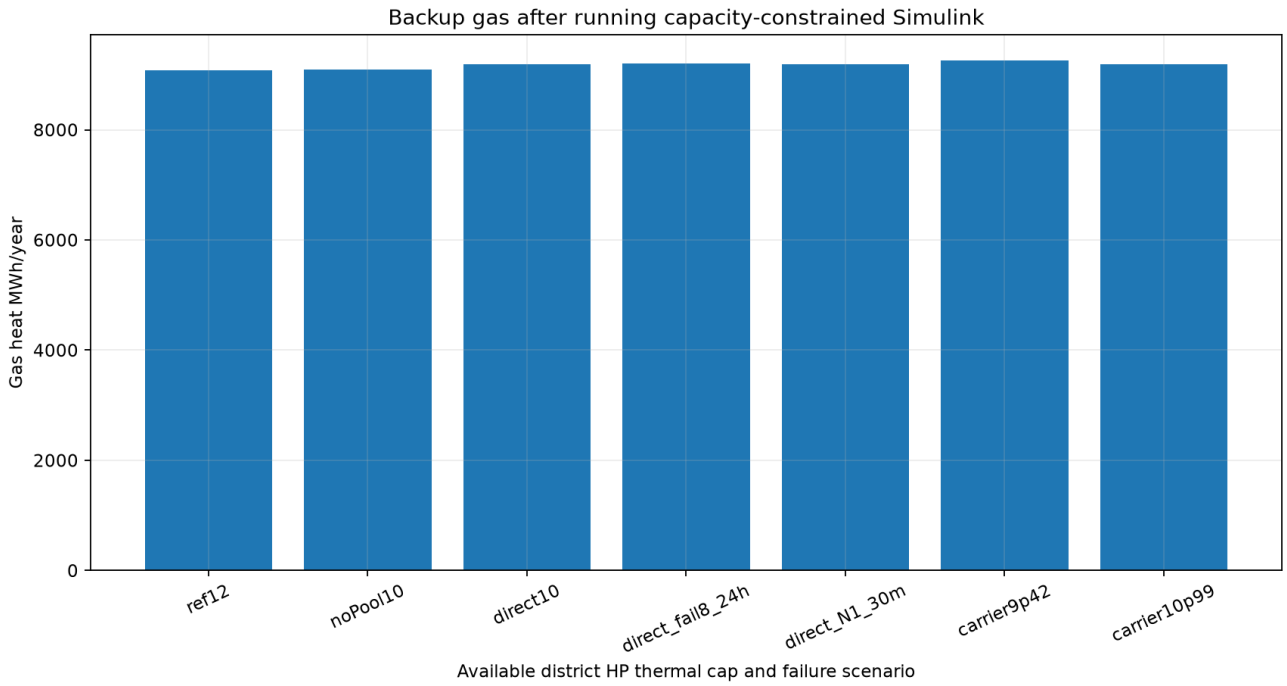
04 load duration



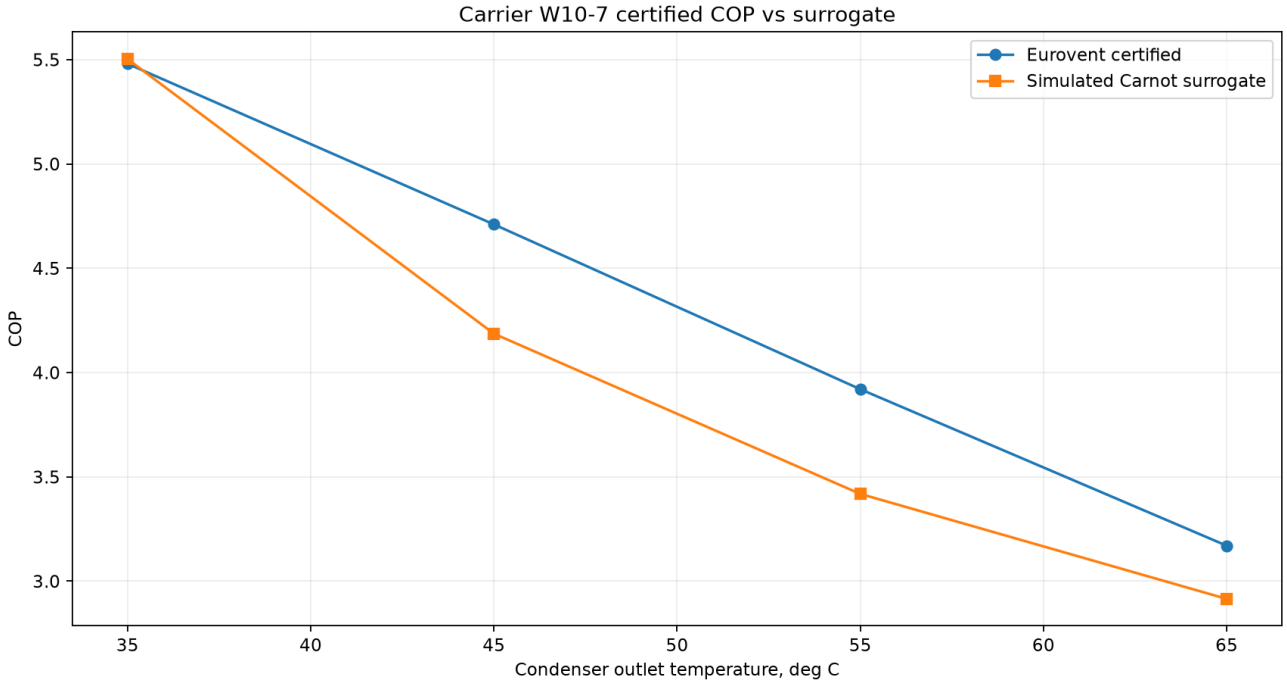
05 IT heat allocation



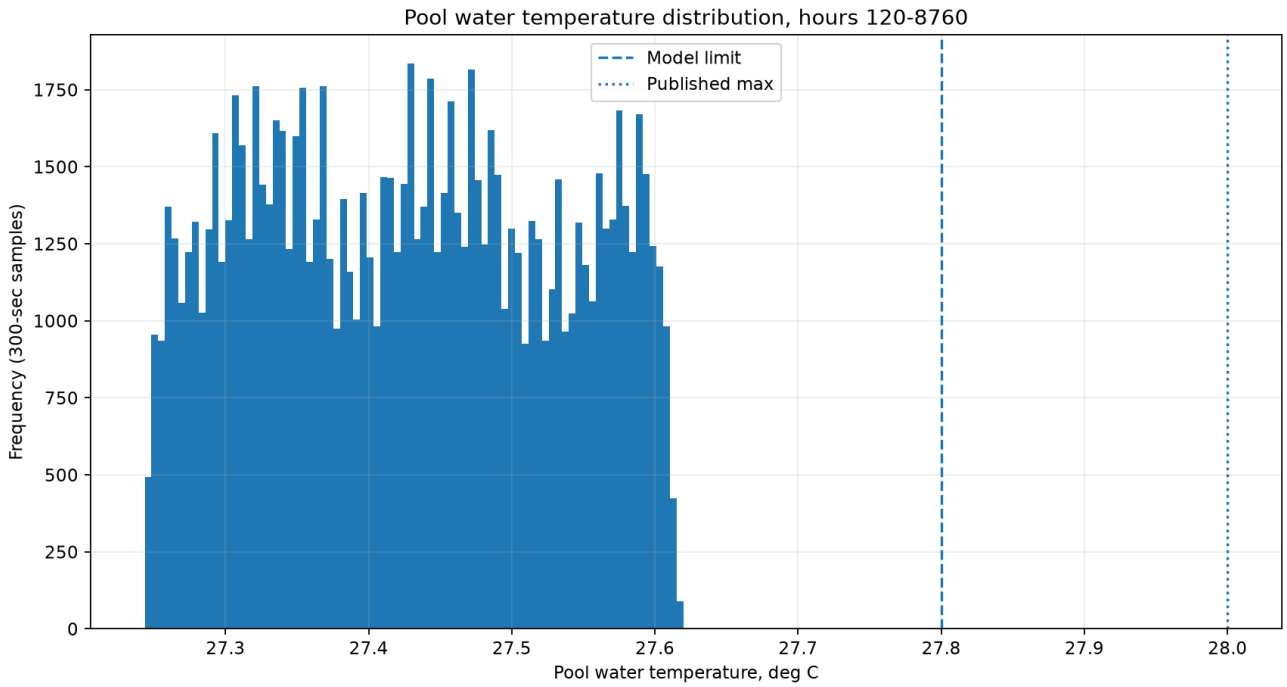
07 HP capacity gas



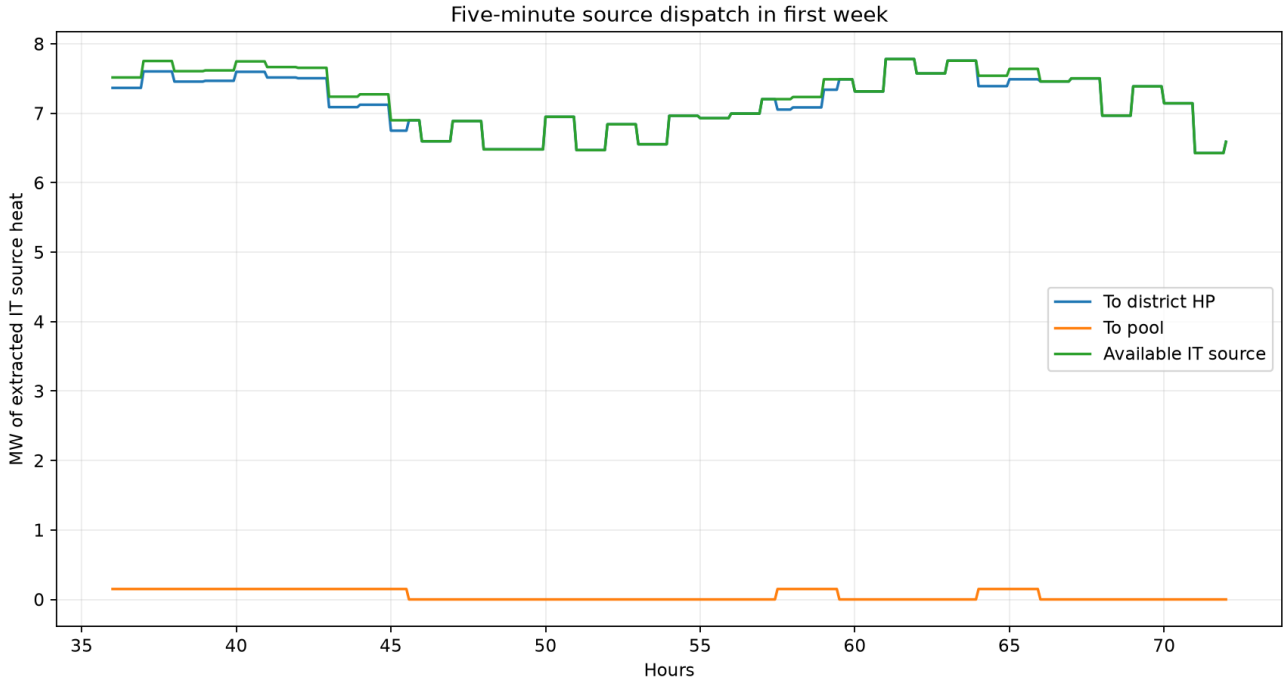
08 COP reference



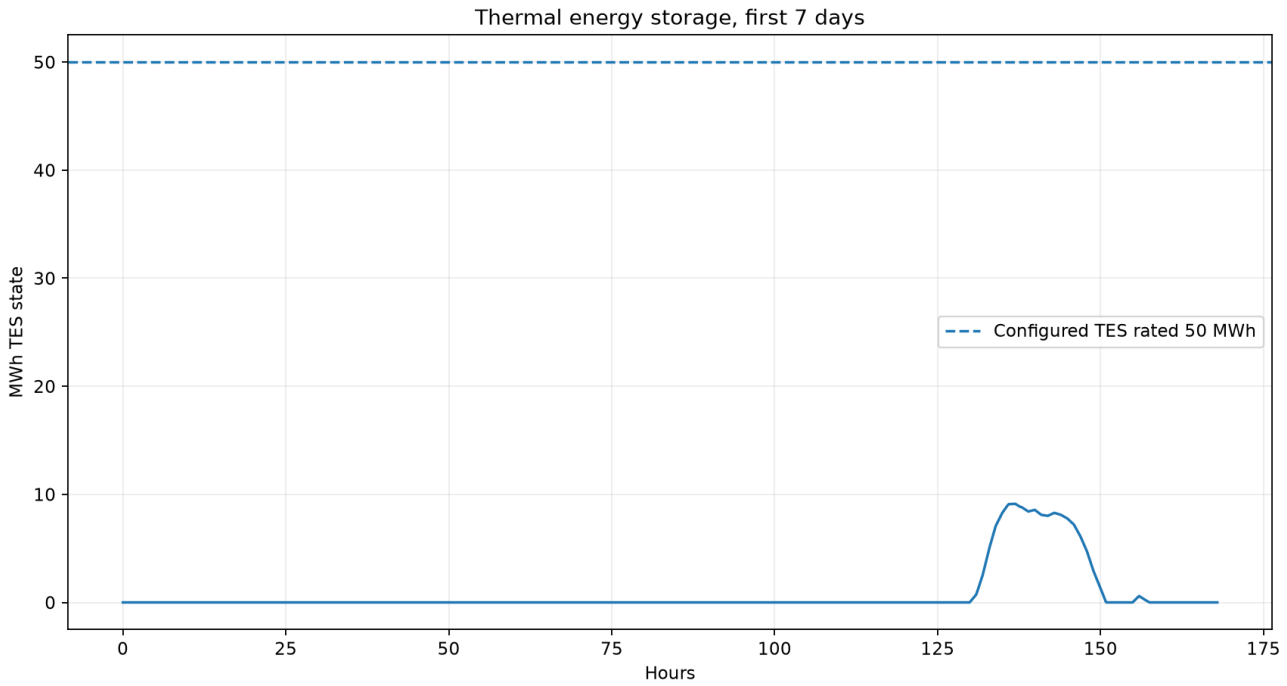
09 pool temperature distribution



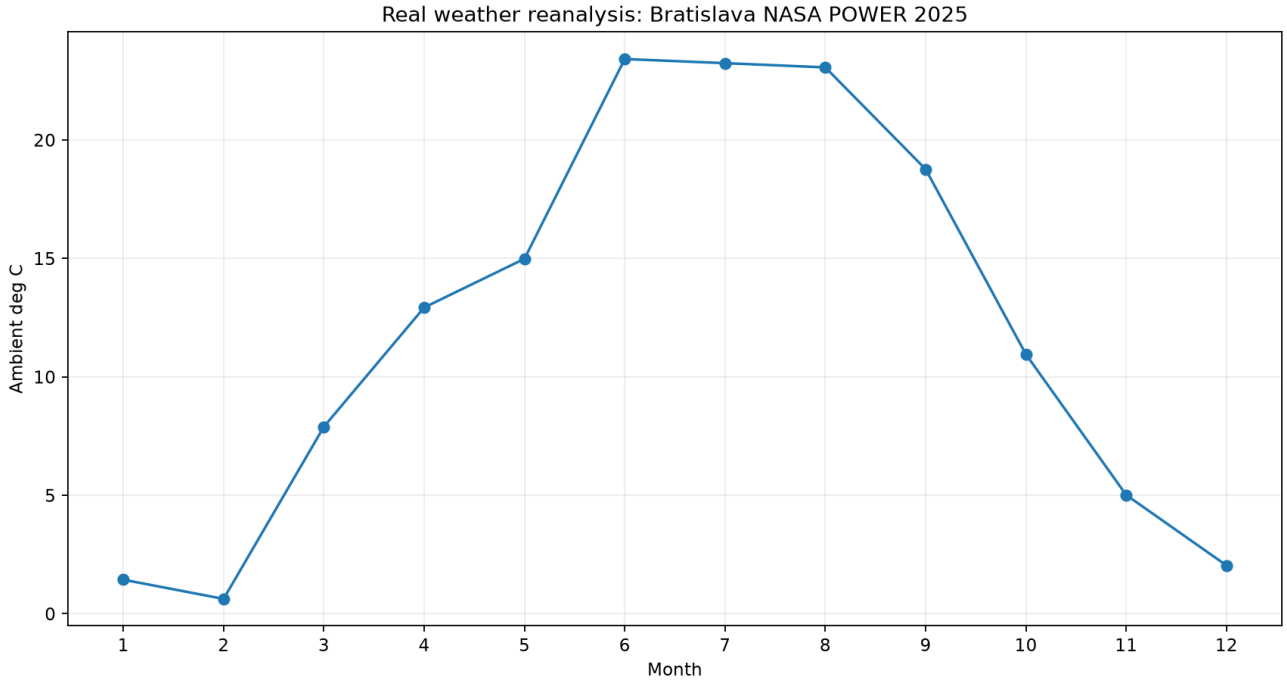
10 recovery dispatch week



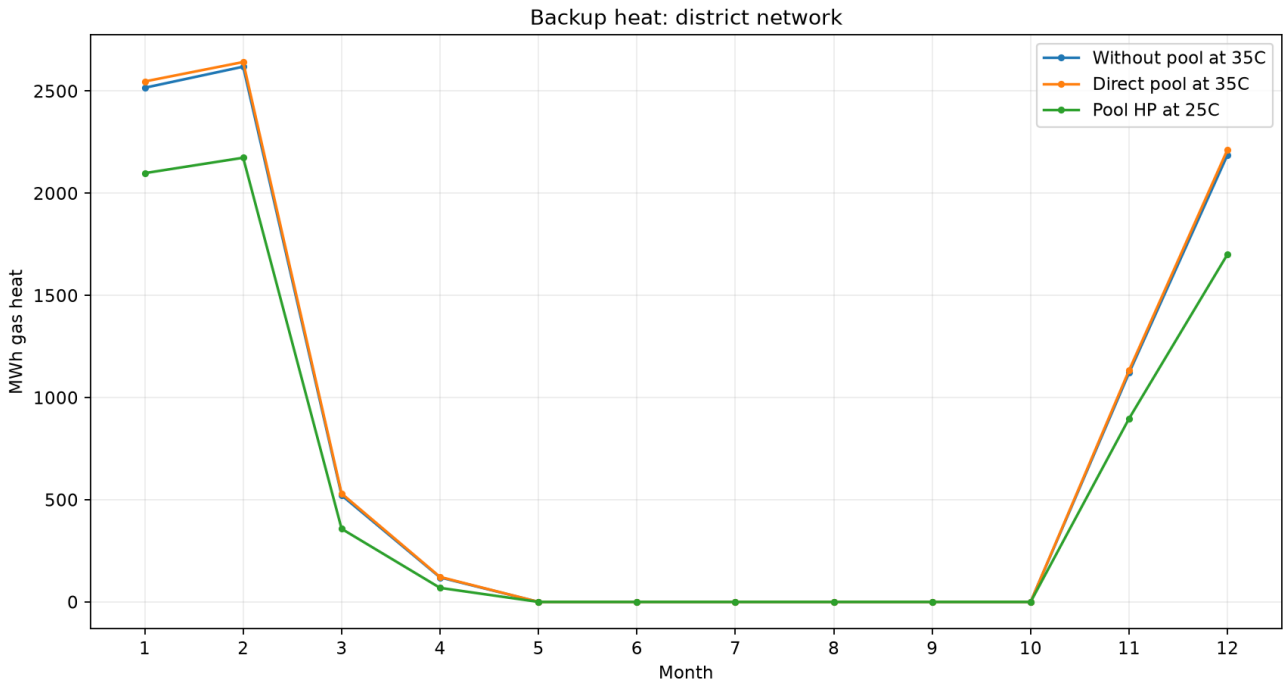
11 thermal storage



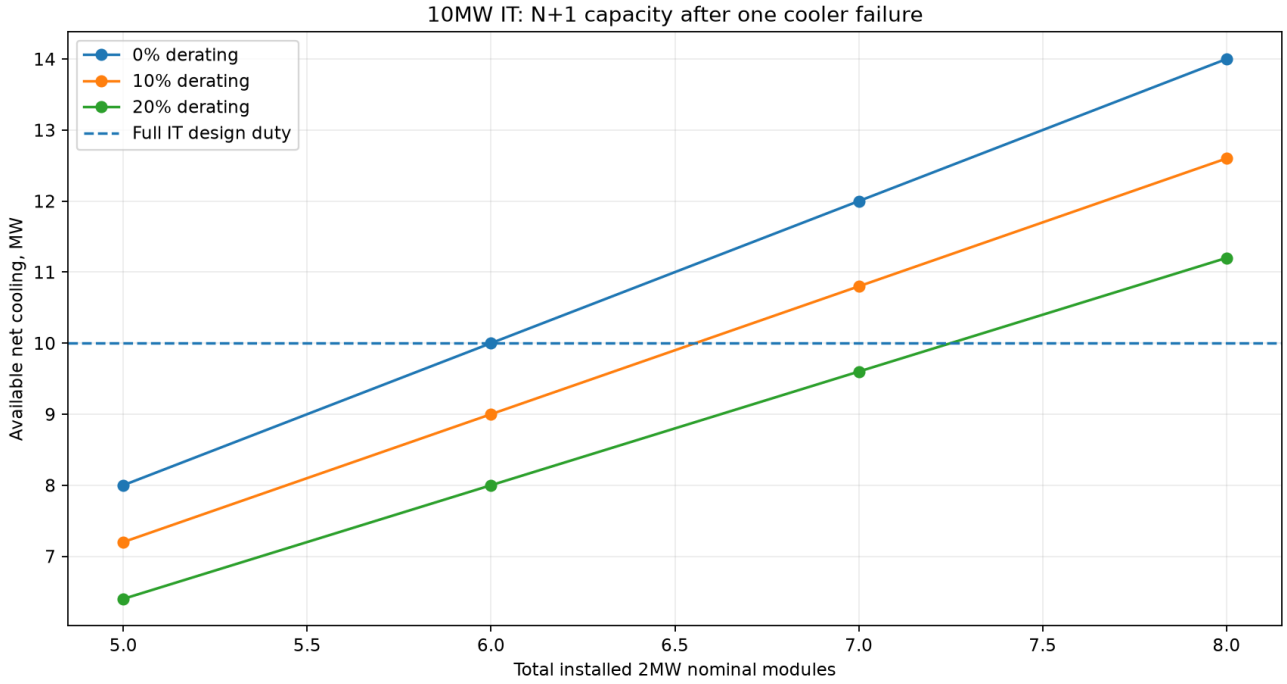
12 weather monthly



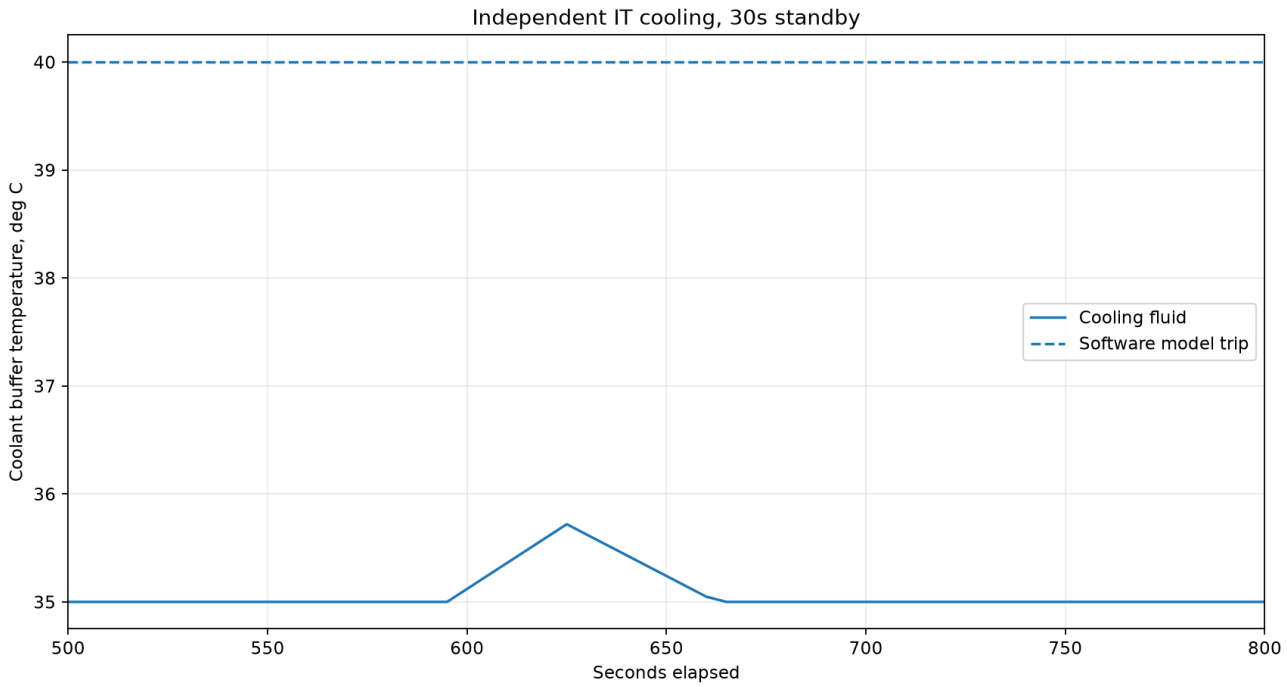
13 gas pool comparison



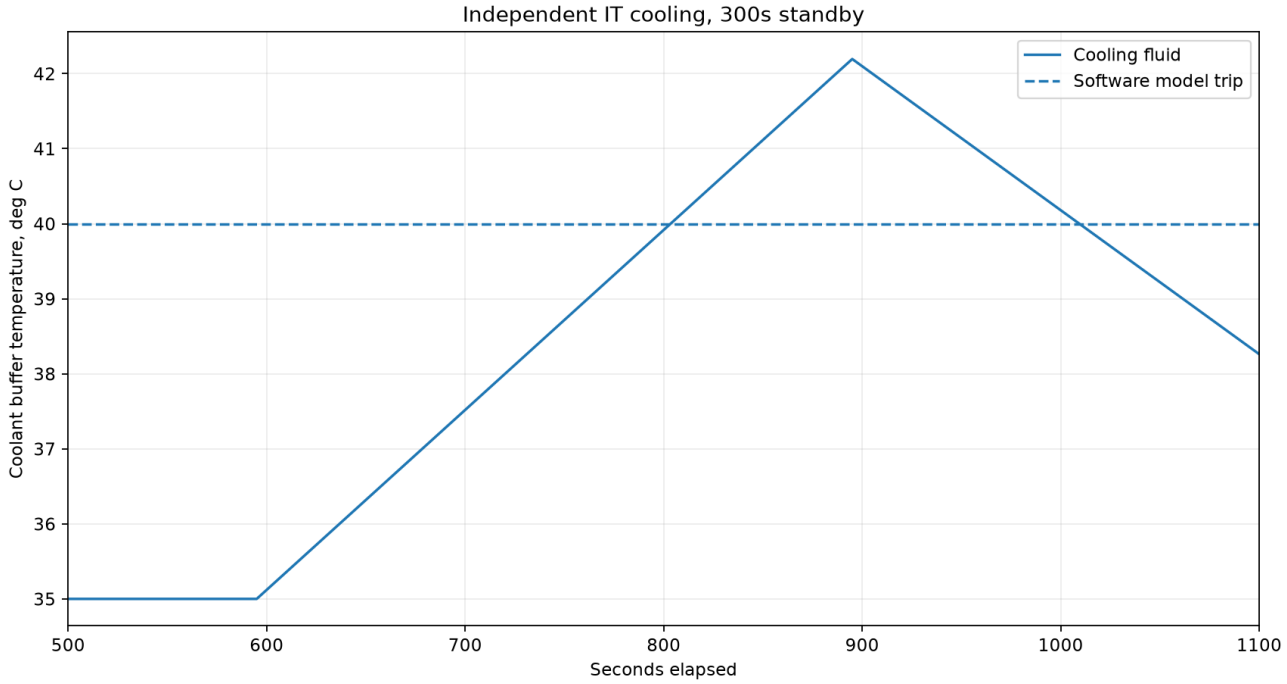
14 independent cooling Nplus1



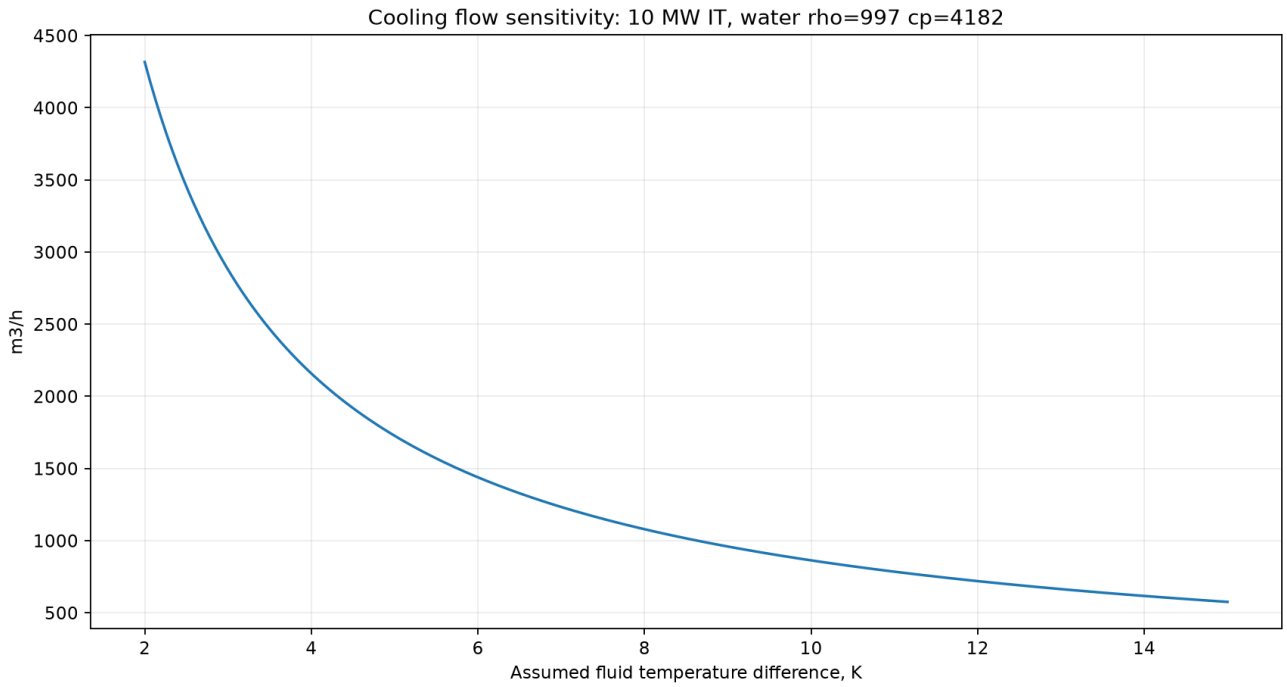
150 cooling fault



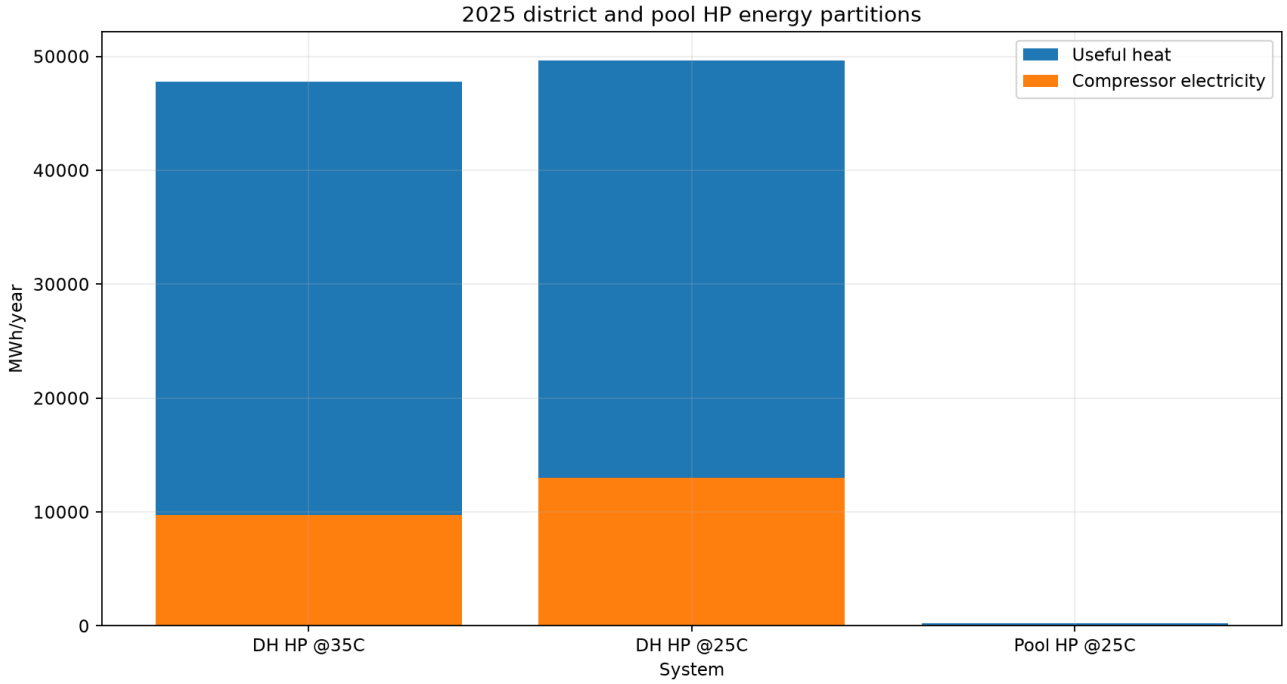
151 cooling fault



17 hydraulic



19 heat pump energy



3. Матрица нормативного соответствия (только проверочный реестр)

56 entries across 22 disciplines. Full licensed standards NOT reviewed; status cannot be marked compliant.

Норма / подсистема	Проверка / подтверждение	Состояние
STD-001 | ISO/IEC 30134-2:2026 | DC Energy	Define DC boundary; E_DC and E_IT measurement categories; annual ratio; Evidence: Meter SLD + calibrated E_DC and E_IT records	P0 | NOT_METERED
STD-002 | EU 2024/1364 Annex II III | DC Energy	Determine installation reporting applicability and annual measurement boundaries; Evidence: Eligibility memo + year metering statement	P0 | NOT_EVALUATED
STD-003 | EN 50600-4-6 | DC Energy	Clarify measuring heat E_REUSE at DC facility boundary; reporting definition; Evidence: Calibrated heat meter with heat transfer diagram	P0 | NOT_METERED
STD-004 | ISO/IEC 30134-9:2022 | DC Energy	Measure site water consumption normalized by IT energy; Evidence: Water submeter + calendar kWh IT	P1 | NOT_METERED
STD-005 | ISO 50001:2018 | DC Energy	Define baseline EnPIs, targets, monitoring and continuous improvement; Evidence: Energy policy; audit; owner KPI reviews	P1 | NOT_IMPLEMENTED
STD-006 | ISO 50002-1:2025 | DC Energy	Document audit boundaries, traceable data and findings; Evidence: Signed audit plan and measurement log	P1 | NOT_IMPLEMENTED
STD-007 | EU Energy Efficiency Directive 2023/1791 | DC Energy	Assess thresholds, permitting, heat reuse and national transposition; Evidence: Legal applicability memo	P0 | NOT_ASSESSED
STD-008 | ISO/IEC 30134 series | DC Energy	Determine required REF, CUE, energy export metrics; Evidence: KPI inventory with applicability	P2 | PARTIALLY_MODELED
STD-009 | EN 50600 series | Data Center	Select facility availability, redundancy and lifecycle documentation; Evidence: Site architecture, basis of design, review	P0 | NOT_VERIFIED
STD-010 | ASHRAE liquid cooling W classes | IT Cooling	Determine actual IT supply fluid limits from hardware vendor; Evidence: IT hardware coolant specification and thermal trip limits	P0 | NOT_VERIFIED
STD-011 | ASHRAE TC 9.9 | IT Cooling	Confirm safe supply and return, moisture and transients; Evidence: IT vendor acceptance report + BMS trends	P0 | MODEL_ONLY
STD-012 | TIA-942 series | IT Cooling	Assess applicability of redundant topology and system classification; Evidence: Site topology, BOD, N+1 failure proof	P1 | APPLICABILITY_UNKNOWN
STD-013 | EN 50600 availability | IT Cooling	Show no single failure removes IT required cooling; Evidence: FMEA + commissioning fault injection	P0 | SIMULATED_ONLY
STD-014 | EN 378-1 family | IT Cooling	Select refrigerant classification, machine room, release safety; Evidence: OEM refrigerant safety assessment	P0 | NOT_SELECTED
STD-015 | STN EN 14511-2:2022 | Heat Pump	Validate COP and heat output at selected water temperatures; Evidence: OEM full- and partial-load heat map	P0 | NOT_CERTIFIED_FOR_SITE
STD-016 | EN 14511-3:2022 | Heat Pump	Validate heating capacity, compressor draw and test tolerances; Evidence: OEM lab test and independent rating	P0 | NOT_VERIFIED
STD-017 | EN 14825 | Heat Pump	Avoid treating constant COP as seasonal SCOP; Evidence: Manufacturer PLR test data plus weather hours	P0 | NOT_AVAILABLE
STD-018 | EN 378-2 family | Heat Pump	Assess location, refrigerant leak, shutdown and ventilation; Evidence: Refrigerant risk assessment, OEM drawings	P0 | NOT_VERIFIED
STD-019 | Regulation (EU) 2024/573 | Heat Pump	Identify applicable gas phase-outs, labelling and training; Evidence: Chosen refrigerant register and controls	P1 | DEVICE_NOT_SELECTED
STD-020 | EU Ecodesign rules | Heat Pump	Determine applicable ecodesign scope and declaration; Evidence: OEM DoC and technical file	P1 | APPLICABILITY_UNKNOWN
STD-021 | OEM EN14511 selected maps | Heat Pump	Get Qheat,Qevap,Welec,COP at 25/20 and 35/30 to 50/70C; Evidence: Signed Carrier/Oilon selection output	P0 | MISSING
STD-022 | STN EN 15288-1+A1:2024 | Pool Design	Classify swimming pool, verify actual geometry/circulation/safety; Evidence: Licensed norm text + signed design statement	P0 | FULL_TEXT_NOT_ACQUIRED
STD-023 | STN EN 15288-2:2019 | Pool Operation	Supervision, water quality, operational procedures; Evidence: Approved operation manual and risk assessment	P0 | NOT_VERIFIED
STD-024 | STN EN 13451-1+A1:2024 | Pool Equipment	Check circulation fittings, suction and relevant machine guards; Evidence: Manufacturer datasheets and pool safety report	P0 | NOT_VERIFIED
STD-025 | Vyhláška 308/2012 Z. z. §8(7) | Pool Law	Confirm effective consolidated law and actual pool category; published swimming max 28C; Evidence: Legal memo; instrumented overtemp shutoff test	P0 | PUBLIC_TEXT_ONLY
STD-026 | Vyhláška 308/2012 Z. z. | Pool Water	Identify actual operating hygiene limits and sampling; Evidence: Licensed water test plan	P0 | NOT_VERIFIED

Норма / подсистема	Проверка / подтверждение	Состояние
STD-027 | ASHRAE HVAC Applications Natatoriums | Pool Heating	Use occupancy/activity cover humidity and VZT instead of fixed assumptions; Evidence: Psychrometric annual airflow/latent balance	P0 | MODEL_SCENARIO_ONLY
STD-028 | STN EN 16798-1 | Pool Ventilation	Confirm indoor humidity ventilation and design comfort; Evidence: Independent hall HVAC model and air measurements	P1 | NOT_VERIFIED
STD-029 | EU pool chemical rules | Pool Equipment	Verify titanium/gasket chlorine and salt exposure limits; Evidence: OEM materials certification and water chemistry	P0 | NOT_SPECIFIED
STD-030 | OEM selected exchanger | HX	Prove real 150kW including fouling and pressure drop; Evidence: Manufacturer thermal sizing report	P0 | MODEL_U_ASSUMPTION
STD-031 | PED 2014/68/EU | HX	Assess assembly classification, pressure rating and conformity; Evidence: Nameplates/DoC/PED category evidence	P0 | NOT_ASSESSED
STD-032 | Water chemistry and potable-water separation | HX	Document independent IT and pool loops, backflow prevention; Evidence: Approved P&ID and isolation logic	P0 | NOT_DESIGNED
STD-033 | STN EN 12828+A1:2014 | Hydraulics	Size flows, expansion, pumps, valves and pressure safety; Evidence: Calculated P&ID with head curves and expansion	P0 | SCREENING_ONLY
STD-034 | EN 13941 | Hydraulics	Determine pipe installation scope and buried district network; Evidence: Licensed pipe system specification	P1 | APPLICABILITY_UNKNOWN
STD-035 | Pump OEM curves | Hydraulics	Ensure hydraulics at all operating and emergency points; Evidence: Q-H + efficiencies, NPSH calculation	P0 | ASSUMPTION_ONLY
STD-036 | OEM TES rated energy and losses | Thermal Storage	Use measured rather than lumped TES efficiency and loss; Evidence: Storage design and commissioning curves	P1 | MODEL_ONLY
STD-037 | Cooling makeup and blowdown | Water	Quantify supply water, wastewater, biocide and disposal; Evidence: Seasonal water balance + permits	P1 | MISSING
STD-038 | IEC 60364 series | Electrical	Verify circuit sizing, protection, RCD, earthing and selectivity; Evidence: Electrical SLD, protection coordination	P0 | NO_SLD
STD-039 | IEC 61439 series | Electrical	Verify switchboard rating and declaration; Evidence: OEM board documentation and inspection	P1 | NOT_SELECTED
STD-040 | IEC 60204-1 | Electrical	Identify machines and emergency stops; Evidence: OEM conformity and wiring tests	P1 | APPLICABILITY_UNKNOWN
STD-041 | UPS + genset manufacturer | Electrical	End-to-end failover and reserve fuel / autonomy; Evidence: UPS loaded test, generator test	P0 | NOT_VERIFIED
STD-042 | IEC 61508 | Control	Determine required safety function integrity from risk assessment; Evidence: Safety requirements specification, proof test	P0 | NEEDS_HAZARD_ASSESSMENT
STD-043 | IEC 62443-3-3 | Control	Zone/conduit segregation, access, logging and patching; Evidence: Threat model and OT security test	P1 | NOT_DESIGNED
STD-044 | Independent overtemperature trip | Control	Independent sensor/cutout with measured uncertainty and overrun; Evidence: Hardware-in-loop trip test	P0 | ONLY_SOFTWARE_SIMULATION
STD-045 | Flow failure/low pressure trip | Control	Fail-safe flow proof from redundant sensors and real server vendor; Evidence: HIL trip to safe shutdown with UPS	P0 | MODEL_TRIP_ONLY
STD-046 | Heat meter standard EN 1434 series | Metering	Calibrate matched flow and temperature sensors and logging intervals; Evidence: Meter certificates, boundary drawing	P0 | NO_METER
STD-047 | ISO/IEC 30134-2:2026 | Metering	Track E_IT/E_DC on same 12mo period without double count; Evidence: Meter calibration, gap flags, billing crosschecks	P0 | UNAVAILABLE
STD-048 | SI units ISO 80000 | Measurement	Require units, valid ranges, sensor uncertainties for every field; Evidence: Signal register and dimensional unit tests	P0 | PARTIAL
STD-049 | ISO 14001:2026 | Environmental	Assess water refrigerant impacts, reporting, waste; Evidence: Environmental impact and refrigerant leakage audit	P2 | NOT_APPLIED
STD-050 | ISO 14064 series | Environmental	Select market/location emissions and official factors; Evidence: GHG scope and third party factors	P1 | MISSING_EMISSION_FACTORS
STD-051 | ISO 9001:2026 | Business	Maintain document revision and traceable verification; Evidence: Revision-controlled documents, deviation/CAPA log	P1 | PARTIAL_ARTIFACT_QA
STD-052 | ISO 31000:2018 | Business	Risk register with likelihood/severity and risk owner; Evidence: Signed FMEA + residual risk reviews	P1 | FMEA_CREATED_NOT_SIGNED
STD-053 | Contracted electricity and gas | Business	Actual tariff and demand charges, not Eurostat mean; Evidence: Signed energy contracts and site budgets	P1 | BENCHMARK_ONLY

Норма / подсистема	Проверка / подтверждение	Состояние
STD-054 | Commissioning/fat/sat | Operations	Run testable independent operational failover and overtemp proof; Evidence: Site FAT/SAT witnessed signed reports	P0 | NO_PHYSICAL_TESTS
STD-055 | Service and calibration | Operations	Choose planned shutdown windows and periodic retest; Evidence: Approved maintenance and calibration schedule	P1 | MISSING
STD-056 | BMS KPI traceability | Operations	Record sample interval, clock timezone and sensor provenance; Evidence: Versioned BMS CSV and commissioning logs	P0 | SYNTHETIC_ONLY

4. Каталог юзкейсов и испытаний

PASS означает проверку существующих численных выходных данных, BLOCKED означает отсутствие обязательных измерений или аппаратных испытаний.

Юзкейс	Результат	Исходные условия и ожидание
UC-001 | Yearly energy source allocation	PASS	NASA 2025 weather, 2025 synthetic IT / No double use of available DC heat
UC-002 | 500m3 heating from 16C	PASS	35C source, HX 150kW / Reach thermostat control band and respect 27.8C
UC-003 | 28C statutory screening	PASS	Pool classified as swimming, published rule / No samples exceed 28C
UC-004 | Overtemperature sensor offset	BLOCKED	Sensor +0.3 K bias, thermal afterrun / Independent hardware trip keeps actual T<28C
UC-005 | Pool HP first law	PASS	25C source, assumed COP 4 / Qheat=Qsource+Welec every 5min
UC-006 | District HP first law	PASS	35C source, synthetic Carnot / Qheat=Qsource+Welec every 5min
UC-007 | District gas demand	PASS	Year simulated all 2025 / Gas thermal nonnegative, demand covered
UC-008 | Thermal store range	PASS	50MWh rated TES model / Stored energy within [0,50]
UC-009 | No heat sink scenario	PASS	IT export closed for 24h hours 48-72 / Pool refuses IT heat when zero source; backup DH gas
UC-010 | Full heat-reuse system unavailable	BLOCKED	No pool/district energy buyer / 10MW independent full IT cooling operates
UC-011 | Direct HX thermal approach	BLOCKED	Tsource 35C and pool return 27.5C / Manufacturer delivers 150kW 35/32 vs 27.5/29.5
UC-012 | Night blanket reduced pool heat	PASS	Initial 500m3 model, covered vs uncovered / Cover scenario lower annual demand
UC-013 | Double volume sensitivity	PASS	500m3 vs 1000m3 scenario / Larger pool requires more heat, under assumed UA
UC-014 | Manufacturers thermal map	BLOCKED	25/20 to 50/70 and 35/30 to 50/70 / Actual signed COP and capacity rated at site PLR
UC-015 | Vendor-certified 10MW duty	BLOCKED	Candidate Carrier 61XWH-ZE17 catalog capacity / Nameplate at required water source and sink is adequate
UC-016 | Nameplate limit 10MW	PASS	Simulink 2025, no pool / 10MW cap changes gas demand by <10MWh/year
UC-017 | Module failed 24 hours	PASS	Simulated 10MW pool dispatch, one HP module -2MW / Gas increases while pool heat remains safe
UC-018 | Standby recovered in 30min	PASS	10MW to 8MW fault and back / Gas incremental less than 1MWh vs healthy
UC-019 | One recooler failed, no summer derating	PASS	6x2MW includes spare, 30s / No thermal trip at 10MW IT load
UC-020 | One recooler failed with derating 10%	PASS	7x2MW includes spare, 30s / No thermal trip at 10MW IT load
UC-021 | Delayed standby 300 seconds	PASS	7x2MW at -10%, 10m3 buffer / Thermal trip is detected, NOT falsely declared safe
UC-022 | Permanent inadequate heat rejection	PASS	6x2MW at -10% / Trip detected before claiming safe operation
UC-023 | Second derating case 20%	PASS	7x2MW at -20% / Trip detected before claiming safe operation
UC-024 | Circulation pump lost 10 seconds	PASS	7x2MW at -10% / Immediate flow trip independently of temperature
UC-025 | UPS pump ride through	BLOCKED	Mains failed, 60s generator delay / Pump flow sustained, redundant electrical supply validated
UC-026 | Cooling switchgear single fault	BLOCKED	Critical cooling distribution fault / Power selectivity and N+1 UPS verified
UC-027 | DC energy efficiency reporting	BLOCKED	Annual same boundary metering / Report official PUE not model 1.36
UC-028 | Regulated energy reuse reporting	BLOCKED	One year heat transfer meter / Report E_REUSE/E_DC compliant with regulation
UC-029 | Water resource KPI	BLOCKED	Pool and DC water demands isolated / Calculate WUE with meter and unit
UC-030 | Timestamp and timezone	PASS	2025 UTC data and Bratislava DST local pool hours / No missing/duplicate 8760 hourly sources
UC-031 | Current PUE standard edition	PASS	Reference year 2026 / ISO/IEC 30134-2:2026 present in register
UC-032 | Latest ISO9001 quality edition	PASS	Reference year 2026 / ISO9001:2026 present, not superseded 2015
UC-033 | Pool chlorinated water chemistry	BLOCKED	HX contact pool treated water / Corrosion rating signed by OEM and sampling
UC-034 | Pool water failsafe FAT/SAT	BLOCKED	Inject stuck valve + T offset / Independent hardware trip before 28C
UC-035 | IT cooling all sinks lost	BLOCKED	Close heating load and simulated DC recoolers / Critical cooling hardware bypass start and server flow maintained

Юзкейс	Результат	Исходные условия и ожидание
UC-036 | ASHRAE W class selection	BLOCKED	Facility coolant supply unknown / Choose compatible W17/W27/W32/W40/W45 or W+ from actual IT design

5. Следующие доказательства для прохождения аудита

Необходимо получить годовые E_IT и E_DC со счётчиков, отдельный E_REUSE на границе DC, подписанные заводские температурные карты HP и низкотемпературного HX, действующие STN с полной проверкой текста, гидравлическую P&ID, SLD/UPS, протоколы аппаратного испытания аварийных функций и подпись инженера. Симуляционный PASS не равен соответствию стандарту.

Source pages: https://www.iso.org/standard/30134-2 ; https://eur-lex.europa.eu/eli/reg_del/2024/1364/oj ; https://normy.normoff.gov.sk/norma/139372/?lang=en ; https://www.ashrae.org/technical-resources/topics-and-initiatives/ai-data-center/introduction-and-purpose